In the footsteps of van der Waals — Solution

W22 (2022) — English translation

A1 — 0.40 pts

Write down an expression for finding the optimal value a_0 through the sums (or averages) of values T, V and their combinations.

We will solve the problem formally, express the temperature from the model function and write down the sum S that needs to be minimized.

$$T = \frac{(p + a_0)\,V}{R}$$

$$S = \sum_n \left(T_i - \frac{pV_i}{R} - \frac{a_0 V_i}{R}\right)^2$$

Differentiate:

$$\frac{\partial S}{\partial a_0} = \sum_n 2\left(T_i - \frac{(p + a_0)\,V_i}{R}\right)\left(-\frac{V_i}{R}\right) = 0$$

Transform (сократив on ”-2”):

$$\frac{1}{R}\sum_n T_i V_i - \frac{p + a_0}{R^2}\sum_n V_i^2 = 0$$

and further

$$p + a_0 = \frac{R\sum_n T_i V_i}{\sum_n V_i^2}$$

We finally get

$$a_0 = \frac{R\sum_n T_i V_i}{\sum_n V_i^2} - p = R\frac{\overline{TV}}{\overline{V^2}} - p$$

Note: The answer could be written almost immediately if you realize that the model formula is linear dependence with one parameter: slope.

A2 — 0.30 pts

Calculate the sums (or averages) you need for the entire data set.

$\sum_n T_i V_i = 6.5903\ \mathrm{K\cdot m^3}, \overline{TV} = 0.31382\ \mathrm{K\cdot m^3}$ $\sum_n V_i^2 = 1.8935\cdot 10^{-5}\ \mathrm{m^6}, \overline{V^2} = 9.0167\cdot 10^{-7}\ \mathrm{m^6}$
Here and below, we save more significant figures to reduce the rounding error of the final answer.

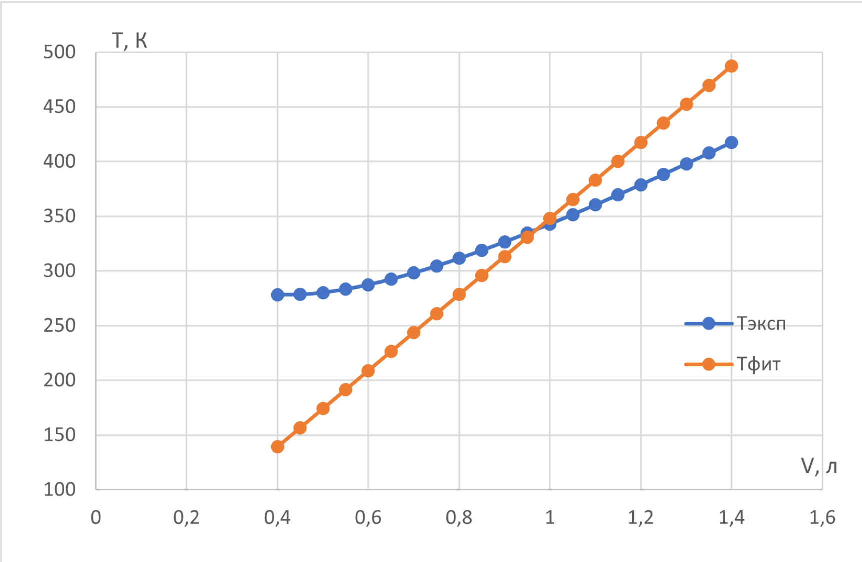
A3 — 0.40 pts

Calculate the value of a_0. Specify it to three significant figures.

$a_0 = 892\ \mathrm{kPa}$

A4 0.40 pts

For each volume value, calculate the temperature value that is obtained as a result of fitting. Plot the initial experimental data and fitting points on one graph. Mark data from different series with different icons. Sign the series.



B1 0.40 pts

Write down an expression for finding the optimal value a_1 through the sums (or averages) of values T, V and their combinations.

Again, we act formally:

$$T = \frac{pV}{R} + \frac{a_1}{R}$$

$$S = \sum_n \left(T_i - \frac{pV_i}{R} - \frac{a_1}{R}\right)^2$$

$$\frac{\partial S}{\partial a_1} = \sum_n 2\left(T_i - \frac{pV_i}{R} - \frac{a_1}{R}\right)\left(-\frac{1}{R}\right) = 0$$

$$\sum_n T_i - \frac{p}{R}\sum_n V_i - \frac{a_1}{R}\sum_n 1 = 0$$

Hence we get окончательный answer:

$$a_1 = \frac{R\sum_n T_i - p\sum_n V_i}{n} = R\overline{T} - p\overline{V}$$

B2 0.30 pts

Calculate the sums (or averages) you need for the entire data set.

$\sum_n T_i = 7009.2$ K$, \overline{T} = 333.771$ K $\sum_n V_i = 0.0189$ m^3$, \overline{V} = 0.0009$ m^3

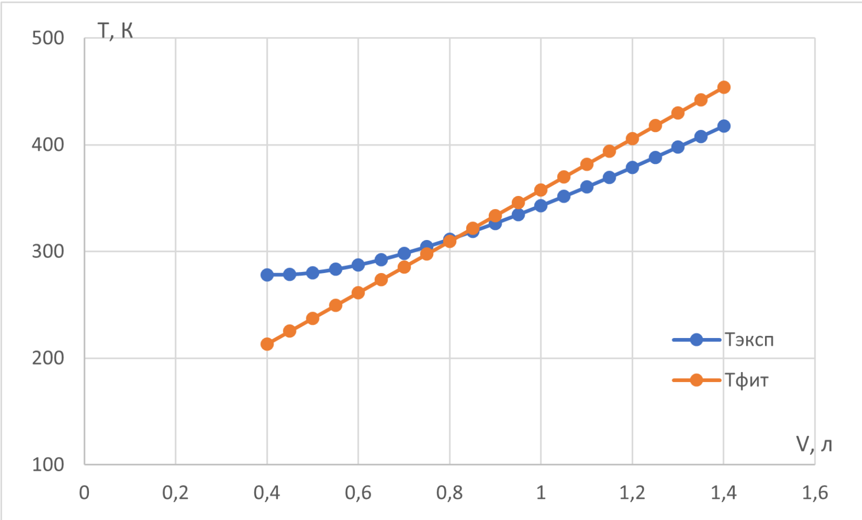
B3 0.40 pts

Calculate the value of a_1. Specify it to three significant figures.

$a_1 = 974$ Pa $\cdot$ m^3

B4 0.40 pts

For each volume value, calculate the temperature value that is obtained as a result of fitting. Plot the initial experimental data and fitting points on one graph. Mark data from different series with different icons. Sign the series.



C1 0.40 pts

Write down an expression for finding the optimal value a_2 through the sums (or averages) of values T, V and their combinations.

Again, we act formally:

$$T = \frac{pV}{R} + \frac{a_2}{VR}$$

$$S = \sum_n \left(T_i - \frac{pV_i}{R} - \frac{a_2}{V_i R}\right)^2$$

$$\frac{\partial S}{\partial a_2} = \sum_n 2\left(T_i - \frac{pV_i}{R} - \frac{a_2}{V_i R}\right)\left(-\frac{1}{V_i R}\right) = 0$$

$$-\frac{1}{R}\sum_n \frac{T_i}{V_i} + \frac{p}{R^2}\sum_n 1 + \frac{a_2}{R^2}\sum_n \frac{1}{V_i^2} = 0$$

Hence we finally get:

$$a_2 = \frac{R\sum_n \frac{T_i}{V_i} - pn}{\sum_n \frac{1}{V_i^2}} = \frac{R\overline{\left(\frac{T}{V}\right)} - p}{\overline{\left(\frac{1}{V^2}\right)}}$$

Note: Интутивно one can be бы пойти путем линеаризации $pV^2 + a_2 = RTV$, and by аналогии with part B1 write, that $a_2 = R\overline{TV} - p\overline{V^2}$. This erroneous path and incorrect solution. For calculation such method estimate for a_2 will be "displaced". In C3, we will calculate the values using both methods and see that using linearization results in a higher sum of squared deviations.

C2 0.30 pts

Calculate the sums (or averages) you need for the entire data set.

$\sum_n \frac{T_i}{V_i} = 8489686$ K$/^3$, $\overline{\left(\frac{T}{V}\right)} = 404271$ K$/^3$ $\sum_n \frac{1}{V_i^2} = 39221160$ m^{-6}, $\overline{\left(\frac{1}{V^2}\right)} = 1867674$ m^{-6}

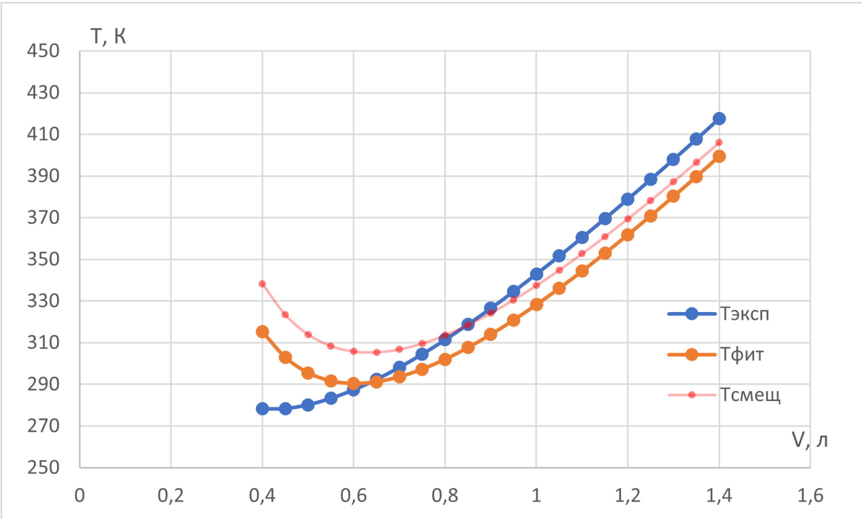
C3 — 0.40 pts

Calculate the value of a_2. Specify it to three significant figures.

$a_2 = 0.728$ Pa · m^6 To calculate a_2^* for the linearized dependence, we have already calculated all the quantities (point A2). Therefore $a_2^* = 0.805$. If you calculate $S(a_2^*) = 8813$ K^2, it will be greater than $S(a_2) = 5478$ K^2.

C4 — 0.40 pts

For each volume value, calculate the temperature value that is obtained as a result of fitting. Plot the initial experimental data and fitting points on one graph. Mark data from different series with different icons. Sign the series.



D1 — 1.50 pts

Write down expressions for finding the optimal values of c and d through the sums (or averages) of values T, V and their combinations.

Let's act again formally:

$$T = \frac{pV}{R} + \frac{C}{RV} + \frac{d}{RV^2}$$

$$S = \sum_n \left(T_i - \frac{pV_i}{R} - \frac{c}{RV_i} - \frac{d}{RV_i^2}\right)^2$$

Differentiate, we get the system of equations:

$$\frac{\partial S}{\partial c} = \sum_n 2\left(T_i - \frac{pV_i}{R} - \frac{c}{RV_i} - \frac{d}{RV_i^2}\right)\left(-\frac{1}{RV_i}\right) = 0$$

$$\frac{\partial S}{\partial d} = \sum_n 2\left(T_i - \frac{pV_i}{R} - \frac{C}{RV_i} - \frac{d}{RV_i^2}\right)\left(-\frac{1}{RV_i^2}\right) = 0$$

Transform system:

$$R\sum_n \frac{T_i}{V_i} - p\sum_n 1 - c\sum_n \frac{1}{V_i^2} - d\sum_n \frac{1}{V_i^3} = 0$$

$$R\sum_n \frac{T_i}{V_i^2} - p\sum_n \frac{1}{V_i} - c\sum_n \frac{1}{V_i^3} - d\sum_n \frac{1}{V_i^4} = 0$$

Transition to mean value and lead to system to convenient form:

$$c\overline{V^{-2}} + d\overline{V^{-3}} = R\overline{T/V} - p$$

14 From which, solve jointly equation, obtain answer:

$$c = \frac{R\left(\overline{T/V^2}\,\overline{V^{-3}} - \overline{T/V}\,\overline{V^{-4}}\right) - p\left(\overline{V^{-1}}\,\overline{V^{-3}} - \overline{V^{-4}}\right)}{\left(\overline{V^{-3}}\right)^2 - \overline{V^{-2}}\,\overline{V^{-4}}}$$

$$d = \frac{R\left(\overline{T/V^2}\,\overline{V^{-2}} - \overline{T/V}\,\overline{V^{-3}}\right) - p\left(\overline{V^{-1}}\,\overline{V^{-2}} - \overline{V^{-3}}\right)}{\overline{V^{-2}}\,\overline{V^{-4}} - \left(\overline{V^{-3}}\right)^2}$$

Note: Similarly part C1, one can be бы lead to модельную function to linear dependence: $cV + d = RTV^2 - pV^3$, and determine coefficient c and d by formula, lead to in introduction to problem. This erroneous path and incorrect solution. For calculation such method estimate for c and d will be "displaced".

D2 — 2.00 pts

Calculate the sums (or averages) you need for the entire data set.

We calculated two values $\overline{T/V}$ and $\overline{V^{-2}}$ in point C2. $\overline{V^{-1}} = 1270.76$ m^{-3} $\overline{V^{-3}} = 3.13694 \cdot 10^9$ m^{-9} $\overline{V^{-4}} = 5.85765 \cdot 10^{12}$ m^{-12} $\overline{T/V^2} = 0.568387 \cdot 10^9$ K/^{6}

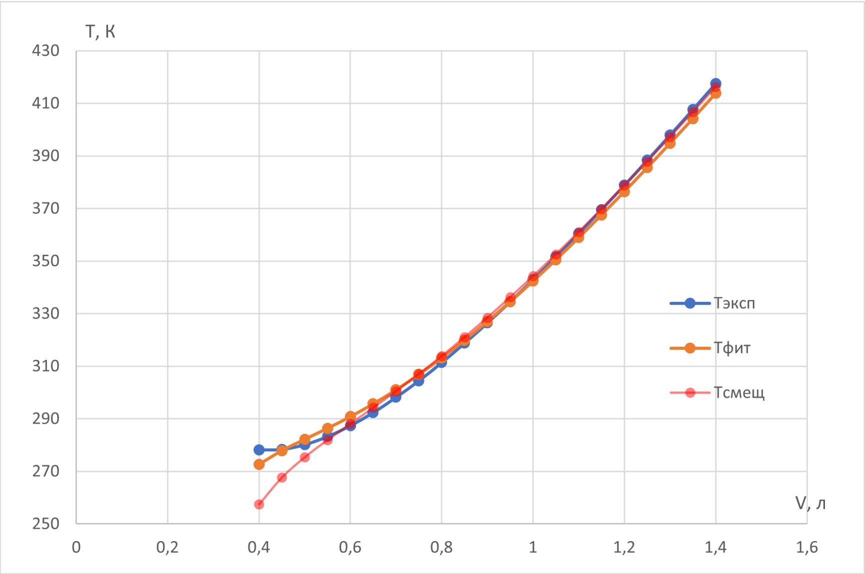
D3 — 1.00 pts

Calculate the values of c and d. Please provide them to three significant figures.

$c = 1.02$ Pa·m^6 $d = -1.73 \cdot 10^{-4}$ Pa·m^9 Offset values: $c^* = 1.08$ Pa·m^6, $d^* = -2.17 \cdot 10^{-4}$ Pa·m^9

D4 — 1.00 pts

For each volume value, calculate the temperature value that is obtained as a result of fitting. Plot the initial experimental data and fitting points on one graph. Mark data from different series with different icons. Sign the series.



E1 **?? pts**

Calculate the values of a and b. Please provide them to three significant figures. Consider the scale of work carried out by Van der Waals.

In this part of the problem, it is necessary to similarly formally minimize the sum over the parameters a and b. The final system of equations for the parameters will look like this:

$$\begin{cases} \alpha_1 a + \alpha_2 b + \alpha_3 ab + \alpha_4 a^2 + \alpha_5 a^2 b = c_1 \\ \beta_1 a + \beta_2 b + \beta_3 ab + \beta_4 b^2 + \beta_5 ab^2 = c_2 \end{cases}$$

Coefficients α_i and β_i are related entirely to the average values calculated in point D2. Those. There would be no additional need to calculate anything. Each of the equations is linear in one of the variables and quadratic in the second. Therefore, the system can be reduced to a fifth-degree equation for one of the variables. An equation of degree five definitely has at least one valid solution.